

Power distribution reconstruction from in-core detector measurements using gaussian process regression

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ABSTRACT

In order to safely and efficiently operate pressurized water reactors, it is essential to monitor neutron flux and power distributions throughout the operation cycle. Most of currently operated reactors have in-core instrumentation systems capable of taking assembly-wise axially integrated flux measurements. However, commercial reactors typically possess few fuel assemblies that contain guide tubes for in-core instrumentation and it is necessary to estimate power values at the remaining fuel assemblies. This work employs gaussian process regression, a non-parametric supervised learning method, to predict the power distribution over the entire core using only measured data. The OpenMC Monte Carlo code was employed to emulate detector signals from instrumented assemblies as well as to produce reference values at non-instrumented assemblies against which the regression method was validated. The mean and maximum relative discrepancies were below 0.8% and 2.5%, respectively, in a scenario with full detector availability. Considering the failure of a single detector, the maximum relative discrepancy did not exceed 5.3%, showing the feasibility of the model as an operation monitoring system capable of real-time reconstruction of the core power distribution.

1. Introduction

Power distribution reconstruction is a crucial procedure in pressurized water reactors, since it enables the evaluation of important operational parameters such as the enthalpy rise factor, the departure from nucleate boiling ratio (DNBR) and the quadrant tilt ratio. Most of the currently operated reactors have in-core instrumentation systems capable of taking assembly-wise axially integrated flux measurements. However, in order to minimize construction and operating costs, commercial reactors possess few fuel assemblies with instrumentation tubes to accommodate neutron detectors. Therefore, it is necessary to estimate power values at the remaining fuel assemblies.

This problem can be addressed from a supervised learning perspective. That is, we wish to learn a function (the power distribution) from a limited data set (the power values at the instrumented fuel assemblies). There is a great diversity of supervised learning algorithms applied to problems in nuclear engineering, such as decision trees, neural networks and support vector machines (Jinia et al., 2024). In this work, we chose to employ the gaussian process regression, a non-parametric bayesian method. This class of methods does not assume a fixed parametric form for the unknown function. Instead, it tries to predict the function itself (and not a set of parameters) from the data.

In the absence of full core experimental power maps, we employed the OpenMC Monte Carlo code to build a model and calculate the power distribution of the Angra 1 reactor. Angra 1 is a two-loop pressurized water reactor located in Brazil. It was commissioned in 1985 and, thanks to its long-term operation (LTO) program, is expected to operate until 2044. This particular reactor was selected to test the proposed methodology because all data required to model the reactor core was available.

Assembly-wise integrated power values were tallied for every depletion step. The power values at the instrumented fuel assemblies were used to train the gaussian process regression model. Once trained, the model was applied to reconstruct the power distribution over the entire core. Finally, the reconstructed power distribution was compared against the Monte Carlo results to validate the proposed methodology. The neutronic simulations performed in this work employed a fully heterogeneous reactor model and continuous-energy dependent cross sections. Therefore, no additional nuclear data processing steps were necessary.

Most power distribution reconstruction methods combine results from a transport or diffusion calculation with in-core detector measurements. Peng et al. (2018) defines a calibration factor for each

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instrumented fuel assembly as the ratio between the measured assembly power and the value calculated by a diffusion code and then estimates this calibration factor for the remaining fuel assemblies using bayesian inference. Power values for non-instrumented assemblies are obtained by multiplying its calibration factor by the corresponding power value calculated by the diffusion simulation. Peng et al. (2019) defines the same calibration factors and reconstructs them using a radial basis function interpolation. Li et al. (2018) compares several reconstruction methods including radial basis function interpolation, support vector machine and distance weighted techniques, employing the same concept of calibration factors. Although commonly used, the calibration factor approach presents a challenge: it is not possible to guarantee that the results of the simulation correspond to the true operational condition of the plant. In practice, various situations not anticipated in the simulations can occur, such as: xenon spatial oscillations, control rod drops and misalignment or failure of fuel assemblies. As demonstrated in Li et al. (2018), an assembly-wise power mapping based on the results of a simulation conducted for one operational condition can perform poorly if the true operational condition is different, even though the calibration factors incorporate the power values measured at the true operational condition.

BEACON, acronym for Best Estimate Analysis of Core Operations, is a core monitoring package developed by Westinghouse and is currently employed at the Angra 1 nuclear power plant. BEACON also utilizes the concept of calibration factor by combining measured data and simulated data obtained via the nodal expansion method (Boyd and Miller, 1996). Subsequently, the power distribution is reconstructed using a surface spline fitting.

Wang et al. (2011) proposes a different approach to the power distribution reconstruction problem. The power distribution is expanded in a set of harmonics of the neutron diffusion equation for multiple reference operational conditions and a data-library is generated. When the power values are measured by the in-core instrumentation system, the two closest reference cases are searched in the data-base and interpolated to reconstruct the measured power distribution.

2. Angra 1 description

2.1. Reactor overview

Angra 1 is a nuclear power plant located in the Almirante Álvaro Alberto Nuclear Power Station (CNAAB), Angra dos Reis city, state of Rio de Janeiro, Brazil. The unit comprises a closed-cycle light water moderated and cooled reactor designed to produce 1876 MWt and 640 MWe (Demas and Somerville, 1976), substantially contributing to Brazil's clean energy mix.

The core of Angra 1 consists of 121 fuel assemblies. Each fuel assembly contains 264 fuel rods and 25 guide tubes arranged in a 16×16 square lattice. 24 out of the 25 guide tubes can accommodate control or burnable poison rods and the central guide tube can accommodate an instrumentation tube. The model used in this work corresponds to the first operation cycle of Angra 1. Therefore, the fuel is composed solely by uranium dioxide with three possible enrichment levels: 2.1%, 2.6% and 3.1%. Fig. 1 shows a schematic representation of the Angra 1 core inspired by the beautiful figures from Horelik et al. (2013).

2.2. In-core instrumentation

The goal of a reactor's instrumentation system is to map power and output coolant temperature distributions to ensure an operation within the plant's safety margins and an efficient utilization of the nuclear fuel. Angra 1 in-core instrumentation system is comprised by mobile neutron detectors and fixed thermocouples.

The central position of each fuel assembly (more precisely, the northwest position among the four innermost positions) is occupied by an empty guide tube. In 36 of the 121 fuel assemblies, the guide tube

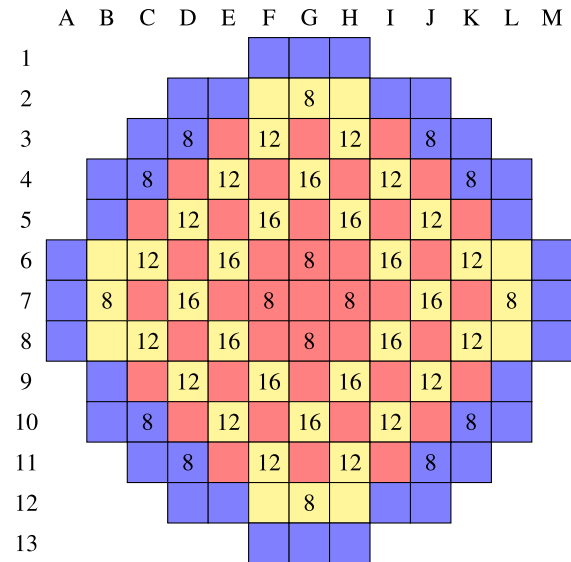


Fig. 1. Angra 1 core cross section view. The colors represent different enrichment levels: pink 2.1%, yellow 2.6% and purple 3.1%. The numbers refer to the quantity of burnable poison rods in each fuel assembly.

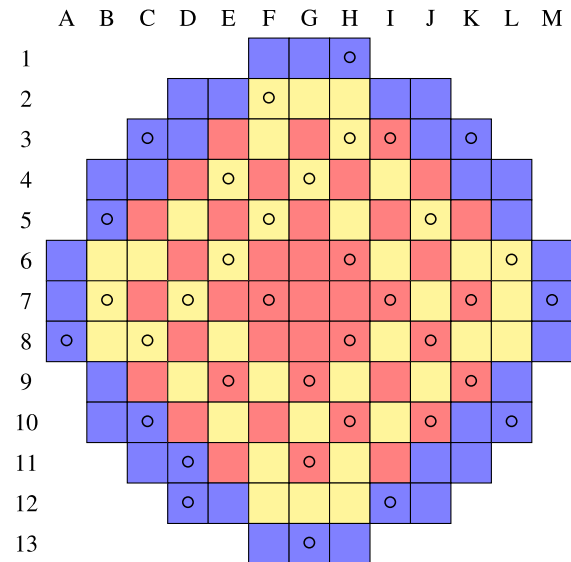


Fig. 2. Instrumented fuel assembly positions in the Angra 1 core.

contains an instrumentation tube that can be accessed by the in-core mobile neutron detectors. The positions of the instrumented assemblies are shown in Fig. 2.

The mobile neutron detectors consist of fission chambers with triuranium octoxide (U_3O_8). Each detector is connected to a flexible cable that can be inserted into one of the 36 instrumentation tubes. The detectors are inserted through the bottom of the reactor vessel and can reach the top of the fuel assemblies. When not in use, the detectors are stored in a concrete vault to minimize the plant's personnel radiation exposure.

Before the measurement process, the detector is inserted into the core until it reaches the top of the fuel assembly. Measurements are taken when the detector is being pulled back at constant speed. Multiple axial measurements are taken and their sum is proportional to the assembly's power.

In addition to mobile detectors based on fission chambers, employed in Angra 1's internal instrumentation system, there are other types of

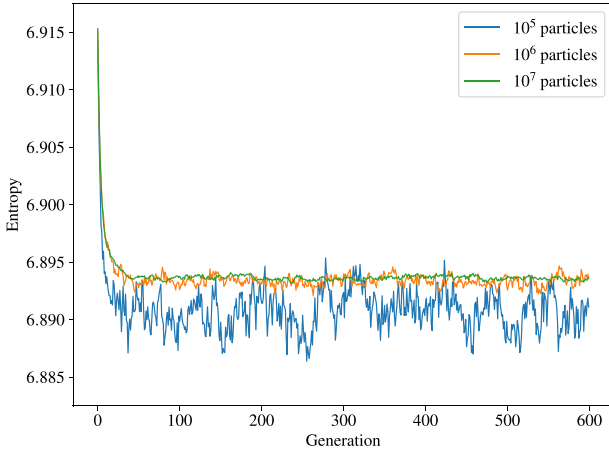


Fig. 3. Shannon Entropy behavior for three simulations of the Angra 1 model with different number of particles per generation.

flux detectors, such as the self-powered neutron detectors (Turcotte, 1974) and the aeroball system (Dias and Silva, 2016).

3. Monte Carlo simulation

3.1. OpenMC Monte Carlo code

OpenMC (Romano et al., 2015) is an open-source neutron and photon Monte Carlo transport code written in C++ and Python. The input files provide OpenMC with the necessary information about the model, the simulation parameters and the quantities the user wish to obtain (tallies, in the OpenMC jargon). The model geometry can be built using constructive solid geometry or CAD representation. OpenMC is capable of performing fixed source or criticality simulations using either continuous-energy or multi-group cross section data. In addition to the transport solver, OpenMC features a built-in depletion solver, which allows the user to analyze the behavior of nuclear reactors throughout the operation cycle (Romano et al., 2021).

3.2. Angra 1 simulation

Aiming to calculate the assembly-wise power distribution, whose values at the instrumented fuel assemblies will be used as training data for the supervised learning method and whose values at the non-instrumented fuel assemblies will be used as reference values to assess the method's performance, we constructed a model of the Angra 1 reactor on the OpenMC Monte Carlo code. The model corresponds to the first cycle of operation and includes fuel rods, burnable poison rods, control rod guide tubes, instrumentation tubes, baffle, core barrel, thermal shield and reactor vessel. Grid spacers and any components above or below the active core were not modeled. We assumed a flat temperature profile, with the coolant temperature fixed at 580.04 K and the fuel temperature fixed at 880.04 K. The cross sections for every nuclide present in the model were obtained from the ENDFB VIII.0 nuclear data library (Brown et al., 2018).

In order to determine the minimum number of particles and inactive generations necessary to converge the source distribution of the eigenvalue problem, we employed the Shannon Entropy metric (Haghighat, 2021). Fig. 3 shows the behavior of the Shannon Entropy for simulations with varying a number of particles considering a 13×13 square mesh over the Angra 1 core that coincides with the fuel assembly lattice in the active core region.

For the scenario of 10^5 particles per generation, the Shannon Entropy exhibits an excessively noisy behavior and it is unclear in which

generation it reaches stability. For the scenario of 10^6 particles per generation, the noisy level reduces considerably and it is possible to verify that the Shannon Entropy reaches stability between generations 100 and 200. For the scenario of 10^7 particles per generation the noisy level is even lower and it is possible to confirm that the Shannon Entropy converges between generations 100 and 200. To adopt a conservative position, we conducted the simulations with 10 million particles per generation, 200 inactive generations and 200 active generations, which guarantees a converged fission source and assembly-integrated power tallies with relative uncertainties below 0.2%.

The model was depleted to a maximum burn-up of 13 MWD/kgU, since this was the maximum burn-up designed for the first cycle of operation of Angra 1 as reported at Demas and Somerville (1976).

4. Gaussian process regression

4.1. Mathematical background

A stochastic gaussian process is any sequence of random variables $\{X_t\}$ indexed by an index $t \in I$ such that any finite subset $X = (X_1, X_2, \dots, X_k)^T$ of its random variables follows a gaussian distribution:

$$X \sim \mathcal{N}(\mu, \Sigma) \quad (1)$$

Gaussian processes can be used to define probability distributions over functions. Consider a set $D = \{x_1, x_2, \dots, x_k\}$ and the set $\mathcal{H}$ of all functions $f : D \rightarrow \mathbb{R}$. Any function $f \in \mathcal{H}$ can be univocally represented by a vector $\mathbf{f} \in \mathbb{R}^k$ such that $\mathbf{f} = (f(x_1), f(x_2), \dots, f(x_k))^T$. If for every point $x_i \in D$ we associate a random variable X_i that can take the values $\{f(x_i) | f \in \mathcal{H}\}$ and that has been extracted from a gaussian process, we will have established a gaussian probability distribution for the random vector $X = (X_1, X_2, \dots, X_k)$ and, consequently, for the functions $f \in \mathcal{H}$:

$$\begin{pmatrix} X_1 \\ \vdots \\ X_k \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \mathbb{E}(X_1) \\ \vdots \\ \mathbb{E}(X_k) \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & \dots & \Sigma_{1k} \\ \vdots & \ddots & \vdots \\ \Sigma_{k1} & \dots & \Sigma_{kk} \end{pmatrix} \right) \quad (2)$$

The mean $\mu = [\mathbb{E}(X_1), \dots, \mathbb{E}(X_k)]^T$ and the covariance matrix $\Sigma = [\Sigma_{ij}]_{i,j=1}^k = [\text{Cov}(X_i, X_j)]_{i,j=1}^k$ of the distribution are defined by a mean function $m : D \rightarrow \mathbb{R}$ and a kernel $\mathcal{K} : D \times D \rightarrow \mathbb{R}$ such that $m(x_i) = \mathbb{E}(X_i)$ and $\mathcal{K}(x_i, x_j) = \Sigma_{ij}$. More compactly, we can write:

$$f(\cdot) \sim \mathcal{GP}(m(\cdot), \mathcal{K}(\cdot, \cdot)) \quad (3)$$

To denote that the function f has been sampled from a gaussian process defined by a mean function m and a kernel $\mathcal{K}$.

Now suppose we observe the function values $\{f(v_i)\}_{i=1}^m$ at the points $\{v_i\}_{i=1}^m$ and wish to estimate the function values at the remaining points $\{w_i\}_{i=1}^n$. In the supervised learning jargon, the observed points are called training points and the unobserved points are called test points. If we define the vector of function values at the training points $\mathbf{f}_v = (f(v_1), \dots, f(v_m))^T$ and the vector of function values at the test points $\mathbf{f}_w = (f(w_1), \dots, f(w_n))^T$, their joint probability distribution will be:

$$\begin{pmatrix} \mathbf{f}_v \\ \mathbf{f}_w \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \mu_v \\ \mu_w \end{pmatrix}, \begin{pmatrix} \Sigma_{vv} & \Sigma_{vw} \\ \Sigma_{wv} & \Sigma_{ww} \end{pmatrix} \right) \quad (4)$$

Where μ_v and μ_w are the corresponding mean vectors and Σ_{vv} , Σ_{vw} , Σ_{wv} and Σ_{ww} the corresponding covariance matrices.

Since the training point values are known, we can compute the conditional probability distribution of the test points given the training data $p(\mathbf{f}_w | \mathbf{f}_v, \mu, \Sigma)$. By the rule of conditioning for gaussian distributions (Murphy, 2022), the probability distribution of the test points will also be gaussian and given by:

$$p(\mathbf{f}_w | \mathbf{f}_v, \mu, \Sigma) = \mathcal{N}(\mu_w + \Sigma_{wv} \Sigma_{vv}^{-1} (\mathbf{f}_v - \mu_v), \Sigma_{ww} - \Sigma_{wv} \Sigma_{vv}^{-1} \Sigma_{vw}) \quad (5)$$

The previous knowledge about the function we wish to reconstruct is inserted into the model through the mean function and the kernel. The mean function provides the model with the best estimate we have for the target function before we have access to any data. The kernel measures the similarity between pairs of points and, therefore, tells the model how close the target function's outputs at those points should be. The meaning of similarity is specified by the structure of the particular kernel used in the problem and is not limited to the euclidean distance between points. Typically, the mean function and the kernel depend on a number of parameters, which we call hyperparameters to highlight that they are not parameters imposed on the target function.

Rarely the initial choice of hyperparameters will define the model that best fits the data. To find the best set of hyperparameters, it is necessary to optimize the model with respect to the training data. A very common approach in other supervised learning techniques is to employ cross-validation (Murphy, 2022). Although cross validation can be used together with the gaussian process regression, the fact that this is a bayesian method based on gaussian probability distributions allows us to optimize the hyperparameters through the maximization of the logarithm of the marginal likelihood:

$$\log p(\mathbf{f}_v | \theta) = -\frac{m}{2} \log 2\pi - \frac{1}{2} \log |\Sigma_{vv}| - \frac{1}{2} (\mathbf{f}_v - \boldsymbol{\mu}_v)^T \Sigma_{vv}^{-1} (\mathbf{f}_v - \boldsymbol{\mu}_v) \quad (6)$$

Where the dependence of Σ_{vv} on the hyperparameters θ is implicit.

Despite being less common than other supervised learning methods, the gaussian process regression presents many advantages over its cousins. Since it is a non-parametric method, it is not necessary to know or assume beforehand a class of functions that satisfactorily describes the problem. In addition, the result of the gaussian process regression is a probability distribution for every test point, a result much richer and more useful than a single value for each test point. Furthermore, a gaussian process regression model is able to incorporate the uncertainty present in the training data. That is, if the training data originate from experimental measurements or any kind of noisy observation of an unknown quantity, the model is able to utilize the uncertainties of these measurements to estimate the uncertainty at the test points more accurately. On the other hand, it is necessary to make structural assumptions about the unknown function (like continuity, differentiability or periodicity) to choose the kernel and the computational cost of inverting the matrix Σ_{vv} grows with $\mathcal{O}(m^3)$ and can become impracticable in time-critical tasks with a large training data set.

4.2. Model implementation

The gaussian process regression was implemented in the Python programming language employing the GPflow (Matthews et al., 2017) library. GPflow is an open-source package written in Python that leverages Tensorflow (Abadi et al., 2015) to build computationally efficient gaussian process regression models.

In order to assign a position to each fuel assembly, we indexed them by the number of the line and the number of the column they occupy in the active core, starting from the upper left corner and ending in the lower right corner (exactly like a matrix). However, to exploit the core symmetry and the resulting symmetry in the power distribution, we shifted the origin of the coordinate system to the center of the core and performed a transformation to polar coordinates. More precisely, for each assembly position (x_i, y_i) , we defined the centralized cartesian coordinates $\tilde{x}_i = x_i - 7$ and $\tilde{y}_i = y_i - 7$ and the centralized polar coordinates:

$$r_i = \sqrt{\tilde{x}_i^2 + \tilde{y}_i^2} \quad \theta_i = \arctan\left(\frac{\tilde{y}_i}{\tilde{x}_i}\right) \quad (7)$$

The coordinates (r_i, θ_i) of the fuel assemblies define the training and test points of the gaussian process.

Table 1

Initial values of the hyperparameters present in Eq. (8).

σ_θ^2	l_θ^2	p	σ_r^2	l_r^2	α
0.1	0.1	$\pi/4$	0.5	0.5	0.5

The mean function was chosen simply as a constant equal to the average of the power values of the instrumented fuel assemblies. The kernel was designed as the product of a rational quadratic kernel that depends only on the radial coordinate r with a periodic kernel that depends only on the angular coordinate θ :

$$\mathcal{K}(\mathbf{r}_i, \mathbf{r}_j) = \sigma_\theta^2 \exp\left[-\frac{2}{l_\theta^2} \sin^2\left(\pi \frac{|\theta_i - \theta_j|}{p}\right)\right] \sigma_r^2 \left(1 + \frac{|r_i - r_j|}{2\alpha l_r^2}\right)^{-\alpha} \quad (8)$$

The initial hyperparameter values (before optimization) are shown in Table 1.

We assumed an uncertainty of 1% for all detector readings. However, this value can be easily adjusted if the true experimental uncertainties are known.

5. Results

5.1. Monte Carlo results

The Angra 1 simulation in OpenMC provided the assembly-wise power distribution for every depletion step. The power values calculated by OpenMC have units of electron-volt per source particle ($eV/particle$). To obtain absolute values in units of Watt (W) it would be necessary to assume the power level of the reactor and normalize every tally with respect to that power level. However, we are only interested in power values relative to the average core power, which is determined by the plant operator. Let q_i be the power value at the i th fuel assembly as calculated by OpenMC. Thus, the core average power is given by:

$$\langle P \rangle = \frac{1}{121} \sum_{i=1}^{121} q_i \quad (9)$$

And the normalized relative power P_i at the i th fuel assembly is given by:

$$P_i = \frac{q_i}{\langle P \rangle} \quad (10)$$

Fig. 4 shows the power distribution for the beginning of cycle (BOC) condition.

5.2. GPR results

The gaussian process regression yields a one-dimensional gaussian probability distribution for the normalized relative power of each fuel assembly ij characterized by a mean μ_{ij} and a variance σ_{ij}^2 . The mean μ_{ij} is the best estimate for the power value of fuel assembly ij and the variance σ_{ij}^2 define the confidence intervals around the mean. Therefore, the most straightforward way to assess the method's performance is to compute how much the reconstructed values μ_{ij} deviate from the reference values P_{ij} obtained from the Monte Carlo simulation. To this end, we define the percent relative discrepancy as:

$$\delta_{ij} = \frac{|P_{ij} - \mu_{ij}|}{P_{ij}} \times 100 \quad (11)$$

Another possible metric to assess the reconstruction performance, considering the method's ability to estimate its own uncertainty, is the standard score Δ_{ij} , defined as:

$$\Delta_{ij} = \frac{|P_{ij} - \mu_{ij}|}{\sigma_{ij}} \quad (12)$$

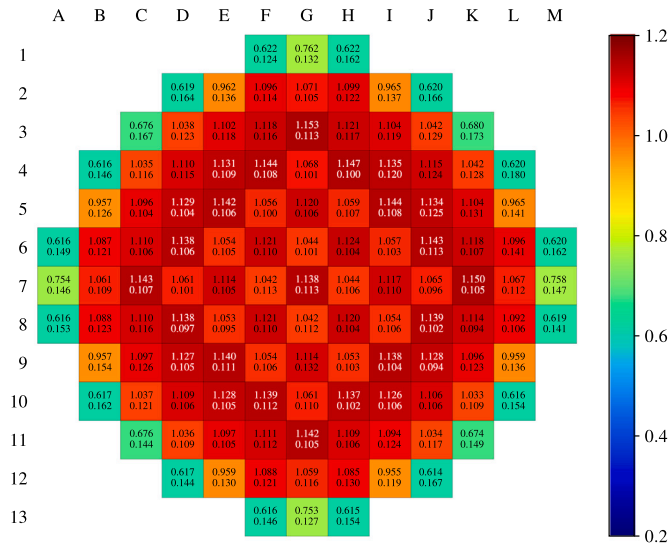


Fig. 4. Assembly-wise power distribution at BOC. Normalized relative power (above) and percent relative uncertainty (below) for each fuel assembly.

Table 2

Summary of the reconstruction results for every depletion step. In the columns, from left to right: burn-up step, mean relative discrepancy, maximum relative discrepancy, mean standard score and maximum standard score.

MWd/KgU	Mean δ	Max δ	Mean Δ	Max Δ
0.00	0.41%	2.07%	0.30	1.82
0.15	0.34%	1.84%	0.27	1.52
1.00	0.40%	2.12%	0.30	1.77
2.00	0.45%	2.21%	0.34	1.99
4.00	0.53%	2.01%	0.41	1.79
6.00	0.47%	2.29%	0.37	2.04
8.00	0.47%	1.77%	0.36	1.57
10.0	0.58%	2.27%	0.44	2.03
12.0	0.71%	2.34%	0.53	2.07
13.0	0.51%	2.28%	0.39	1.37

The standard score is a measure in multiples of the standard deviation of how much the reconstructed values deviates from the reference values. Therefore, it assesses whether the predicted standard deviations are compatible with the observed discrepancies or not. Table 2 presents a summary of the performance metrics for each depletion step.

The overall maximum relative discrepancy and maximum standard score were 2.34% and 2.07, respectively, and both occurred at the 12.0 MWd/kgU depletion step. This worst-case step is highlighted in Table 2 and its detailed results are shown in Fig. 5.

The time required to reconstruct all 10 depletion steps was merely 7.15 s using a 2.8 GHz base-clock and 8-core CPU.

In order to assess the performance of the reconstruction method under a restriction of the training data, we simulated the failure of each in-core detector and evaluated the core average and maximum relative discrepancies for each situation. Essentially, we conducted a leave-one-out cross-validation of our regression model. Fig. 6 shows the result of this analysis for the 12 MWd/kgU burn-up step. That is, we are stressing the model under the previously identified worst-case scenario.

Note that 3 of the 36 detectors produced relative discrepancies exceeding 3%. More precisely, detectors B7, E9 and J10 produced relative discrepancies of 3.31%, 5.07% and 5.29%, respectively. These 3 detectors do not possess symmetric detectors in the other quadrants of the core. Since the kernel used in the regression has a periodic

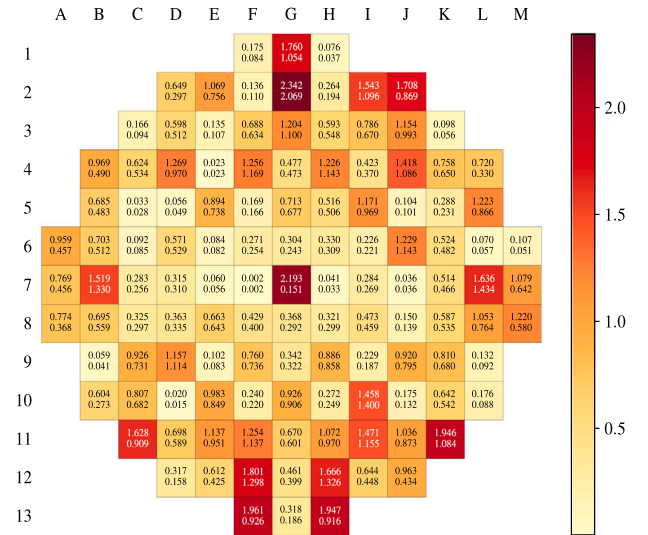


Fig. 5. Detailed reconstruction results for the 12.0 MWd/kgU depletion step. For each fuel assembly, the percent relative discrepancy is shown above and the standard score below. The color scale refer to the percent relative discrepancy values.

component in the angular coordinate θ , it can be expected that the removal of training points without quadrant symmetric points will impact the prediction capability of the method.

6. Summary and conclusion

In a scenario of full detector availability, the proposed gaussian process regression model was able to predict the assembly-wise power distribution in the Angra 1 core with a maximum relative discrepancy of 2.34% and a maximum standard score of 2.07 while the mean relative discrepancy and the mean standard score did not exceed 0.71% and 0.53, respectively. Additionally, the model predicted the power distribution with a maximum relative discrepancy of 5.29% in the event of a single detector failure. These results, combined with a quick computation time, shows that the proposed model can accurately predict the power distribution as well as give reliable estimates of its

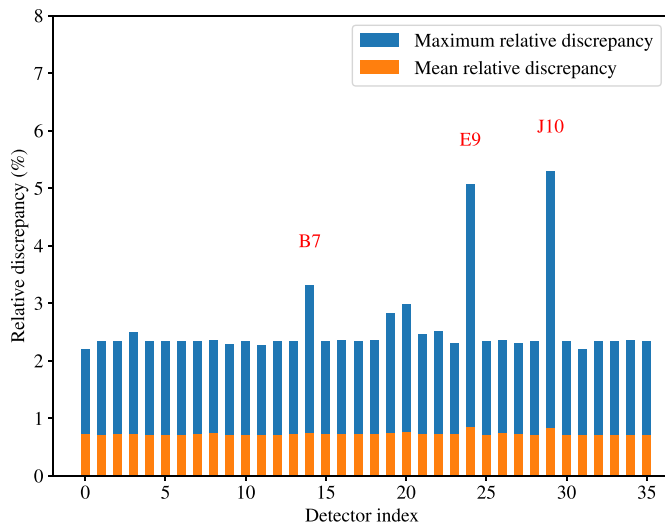


Fig. 6. Mean and maximum average discrepancies considering the failure of each in-core detector at the burn-up step of 12 MWd/kgU. The detectors that when failed produced relative discrepancies larger than 3% are labeled in red.

own uncertainty and demonstrate its feasibility as a real-time power distribution reconstruction method in pressurized water reactors.

In contrast to other approaches found on the literature, the methodology developed in this work depends only on the in-core detector signals. The Monte Carlo simulation was used merely to emulate detector measurements in instrumented assemblies and produced reference values in non-instrumented assemblies. A rigorous comparison with the previously cited studies is not possible, since different reactors were analyzed. However, the average and maximum relative discrepancies obtained from the gaussian process regression are similar to the ones obtained from the other methodologies.

Since the proposed reconstruction method does not depend on simulation results, it potentially can be applied to detect anomalies in the core that disturb the steady-state power distribution, such as xenon spatial oscillations, control rod drop accidents and fuel assembly failures. Nevertheless, further studies are required to confirm this potential, which will likely involve designing a more general kernel.

In addition to the in-core detectors, most commercial reactor's instrumentation systems are comprised by thermocouples that measure the output coolant temperature. However, only a limited number of fuel assemblies are equipped with thermocouples and it is necessary to estimate temperature values for the remaining assemblies, similarly to what happens with the power distribution. Therefore, we also propose as a future study the employment of the gaussian process regression to reconstruct the assembly-wise temperature distribution in pressurized water reactors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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